

GRE Tunnel Configuration

Practical 3

Aim. Build a GRE tunnel between R1 and R3 through R2, then verify routing between their loopback networks.

Topology and addressing

Router	Interface	IP address
R1	FastEthernet0/0	1.1.1.1/24
R1	Loopback0	10.1.1.1/24
R1	Tunnel1	10.0.0.1/24
R2	FastEthernet0/0	1.1.1.2/24
R2	Serial1/0	2.2.2.2/24
R2	Loopback0	20.2.2.1/24
R3	Serial1/0	2.2.2.3/24
R3	Loopback0	30.1.1.1/24
R3	Tunnel1	10.0.0.3/24

Configuration summary

- R1 and R3 use Tunnel1 addresses 10.0.0.1/24 and 10.0.0.3/24. Tunnel endpoints are 1.1.1.1 and 2.2.2.3.
- Static routes through R2 provide underlay reachability. EIGRP AS 10 runs across the tunnel and advertises the loopback networks.
- R1 Tunnel1 MTU: 1300 bytes; TCP MSS adjustment: 1360.

Results

- Tunnel1 reached up/up on both routers; tunnel-address pings succeeded at 100% (5/5) in both directions.
- EIGRP formed a neighbor relationship across Tunnel1, and each router learned the other's loopback route. R2 retained only its connected routes.
- Loopback pings between R1 (10.1.1.1) and R3 (30.1.1.1) succeeded at 100% (5/5) in both directions. R1's tunnel MTU was verified at 1300 bytes.

Note: The R2–R3 connection uses Serial interfaces in this topology, so tunnel endpoints and next hops match the lab interfaces.

Configuration Evidence

Console screenshots from the GRE setup and verification.

Interfaces

R1 interface status

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	1.1.1.1	YES	manual	up	up
Serial1/0	unassigned	YES	unset	administratively down	down
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down
Loopback0	10.1.1.1	YES	manual	up	up

R3 interface status

```
R3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
Serial1/0	unassigned	YES	unset	administratively down	down
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down
Serial1/4	unassigned	YES	unset	administratively down	down
Serial1/5	unassigned	YES	unset	administratively down	down
Serial1/6	unassigned	YES	unset	administratively down	down
Serial1/7	unassigned	YES	unset	administratively down	down

R2 interface status

```
R2#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
Serial1/0	unassigned	YES	unset	administratively down	down
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down

R2 IP configuration and status

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface fastethernet0/0
R2(config-if)#ip address 1.1.1.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface serial1/0
R2(config-if)#ip address 2.2.2.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface loopback0
R2(config-if)#ip address 20.2.2.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#end
R2#
*Oct 7 23:31:06.227: %SYS-5-CONFIG_I: Configured from console by console
R2#
*Oct 7 23:31:28.419: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R2#
*Oct 7 23:31:28.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R2#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	1.1.1.2	YES	manual	up	up
Serial1/0	2.2.2.2	YES	manual	up	up
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down
Loopback0	20.2.2.1	YES	manual	up	up

R3 IP configuration and status

```
R3#enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface serial1/0
R3(config-if)#ip address 2.2.2.3 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface loopback0
R3(config-if)#ip address 30.1.1.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#end
R3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
Serial1/0	2.2.2.3	YES	manual	up	up
Serial1/1	unassigned	YES	unset	administratively down	down
Serial1/2	unassigned	YES	unset	administratively down	down
Serial1/3	unassigned	YES	unset	administratively down	down
Serial1/4	unassigned	YES	unset	administratively down	down
Serial1/5	unassigned	YES	unset	administratively down	down
Serial1/6	unassigned	YES	unset	administratively down	down
Serial1/7	unassigned	YES	unset	administratively down	down
Loopback0	30.1.1.1	YES	manual	up	up

R1 to R2 ping

```
Loopback0 10.1.1.1 YES manual up
R1#ping 1.1.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 24/31/36 ms
R1#
```

Connectivity

R2 to R1 and R3 pings

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/38/64 ms
R2#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/26/36 ms
R2#
R2#ping 2.2.2.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/29/40 ms
R2#
```

R3 to R2 ping

```
*Oct 7 23:33:08.607: %SYS-5-CONFIG_I: Configured from console by console
R3#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/31/44 ms
R3#
```

GRE tunnel

R1 Tunnel1 configuration

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface tunnel1
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#tunnel source fastethernet0/0
R1(config-if)#tunnel destination 2.2.2.3
R1(config-if)#no shutdown
R1(config-if)#end
R1#
*Oct 7 23:36:25.407: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to down
R1#
*Oct 7 23:36:25.895: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R3 Tunnel1 configuration

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface tunnel1
R3(config-if)#ip address 10.0.0.3 255.255.255.0
R3(config-if)#tunnel source serial1/0
R3(config-if)#tunnel destination 1.1.1.1
R3(config-if)#no shutdown
R3(config-if)#end
R3#
*Oct 7 23:37:11.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to down
R3#
*Oct 7 23:37:11.439: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

Static routing

R1 static route

```
R1(config)#ip route 2.0.0.0 255.0.0.0 1.1.1.2
R1(config)#end
R1#
*Oct 7 23:39:09.535: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Oct 7 23:39:14.443: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
R1#
```

R3 static route

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 1.0.0.0 255.0.0.0 2.2.2.2
R3(config)#end
R3#
*Oct 7 23:40:10.067: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
R3#
*Oct 7 23:40:10.367: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

Tunnel test

R1 to R3 tunnel ping

```
R1#ping 10.0.0.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/80/152 ms
R1#
```

R3 to R1 tunnel ping

```
R3#ping 10.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/61/68 ms
R3#
```

EIGRP

R1 EIGRP configuration

```
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router eigrp 10
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#
*Oct 7 23:42:14.471: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R3 EIGRP configuration

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router eigrp 10
R3(config-router)#network 10.0.0.0
R3(config-router)#network 30.0.0.0
R3(config-router)#no auto-summary
R3(config-router)#end
R3#
*Oct 7 23:42:32.415: %DUAL-5-NBRCHANGE: EIGRP-IPv4 10: Neighbor 10.0.0.1 (Tunnel1) is up: new adjacency
*Oct 7 23:42:33.211: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

MTU

Tunnel MTU and configuration verification

```
R1#show ip interface tunnel1
Tunnel1 is up, line protocol is up
Internet address is 10.0.0.1/24
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Multicast reserved groups joined: 224.0.0.38
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Local Proxy ARP is disabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachable are always sent
ICMP mask replies are never sent
IP fast switching is enabled
IP flow switching is disabled
IP CEF switching is enabled
IP CEF switching turbo vector
IP CEF turbo switching turbo vector
Associated unicast routing topologies:
+
R1#show running-config interface tunnel1
+
% Invalid input detected at '^' marker.
R1#show running-config interface tunnel1
Building configuration...
Current configuration : 156 bytes
+
interface Tunnel1
ip address 10.0.0.1 255.255.255.0
ip mtu 1500
ip tcp adjust-mss 1300
tunnel source FastEthernet0/0
tunnel destination 2.2.2.3
end
R1#
```